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ABC-dg:

$$T_{3S/2r} = T_{3S/2S} - T_{2S/2r}$$

$$= \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \\ 0 & 0 \end{bmatrix}$$

对行列式

所对应的

量理解读

正确应该是

$$C_{11} = \cos\theta + \frac{1}{2}\sin\theta$$

$$C_{12} = \sin\theta - \frac{1}{2}\cos\theta$$

$$C_{13} = -\frac{1}{2}$$

$$C_{21} = -\frac{\sqrt{3}}{2}\sin\theta$$

$$C_{22} = \frac{\sqrt{3}}{2}\cos\theta$$

$$C_{23} = -\frac{\sqrt{3}}{2}$$

$$C_{31} = \frac{1}{2}\cos\theta - \frac{1}{2}\sin\theta$$

$$C_{32} = \frac{1}{2}\sin\theta + \frac{1}{2}\cos\theta$$

$$C_{33} = \frac{1}{2}$$

$$\therefore C = \begin{bmatrix} \cos\theta + \frac{1}{2}\sin\theta & \sin\theta - \frac{1}{2}\cos\theta & -\frac{1}{2} \\ -\frac{\sqrt{3}}{2}\sin\theta & \frac{\sqrt{3}}{2}\cos\theta & -\frac{\sqrt{3}}{2} \\ \frac{1}{2}\cos\theta - \frac{1}{2}\sin\theta & \frac{1}{2}\sin\theta + \frac{1}{2}\cos\theta & \frac{1}{2} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



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第1步 (对应A相):

$$C_{11} = 1 \cdot \cos\theta + 0 \cdot \sin\theta + \frac{1}{2} \cdot 0 = \cos\theta$$

$$C_{12} = 1 \cdot (-\sin\theta) + 0 \cdot \cos\theta + \frac{1}{2} \cdot 0 = -\sin\theta$$

$$C_{13} = 1 \cdot 0 + 0 \cdot 0 + \frac{1}{2} \cdot \sqrt{3} = \frac{\sqrt{3}}{2}$$

-B相

$$C_{21} = -\frac{1}{2} \cos\theta + \frac{\sqrt{3}}{2} \sin\theta$$

$$C_{22} = \frac{1}{2} \sin\theta + \frac{\sqrt{3}}{2} \cos\theta$$

$$C_{23} = \frac{1}{2}$$

C相

$$C_{31} = -\frac{1}{2} \cos\theta - \frac{\sqrt{3}}{2} \sin\theta$$

$$C_{32} = \frac{1}{2} \sin\theta - \frac{\sqrt{3}}{2} \cos\theta$$

$$C_{33} = \frac{1}{2}$$

去掉 $\frac{1}{2}$ 系数得

$$\begin{bmatrix} \cos\theta & \cos(\theta - \frac{2\pi}{3}) & \cos(\theta + \frac{2\pi}{3}) \\ -\sin\theta & -\sin(\theta - \frac{2\pi}{3}) & -\sin(\theta + \frac{2\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$





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注: $ABC-dq$ 与 $dq-ABC$
为正交变换

matlab 下的 $A'B'C' - dq'$

$$T_{3/2} = \frac{2}{3} \begin{bmatrix} \sin\theta & \sin(\theta - 2\pi/3) & \sin(\theta + 2\pi/3) \\ \cos\theta & \cos(\theta - 2\pi/3) & \cos(\theta + 2\pi/3) \end{bmatrix}$$



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